

CS6450 · Visual Computing

DYNAPROMPT

Dynamic Test-Time Prompt Tuning

ICLR 2025

Paper by:

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Real-World Example: Intelligent Surveillance System

- ▶ A vision model deployed to monitor public places
- ▶ Initially trained on standard images (clear, daylight, normal conditions)
- ▶ In the real world:
 - Lighting changes (night, fog, rain)
 - New objects or unusual patterns appear
 - Data distribution **continuously shifts**
- ▶ How can the system **learn from new data** while **avoiding incorrect adaptations**?

- ▶ Pretrained models like **CLIP** perform well in zero-shot settings
- ▶ **Performance degrades under distribution shift**
 - Domain shift (e.g. real $\rightarrow$ sketch)
 - Cross-dataset shift (e.g. ImageNet $\rightarrow$ satellite images)
- ▶ Lack of labeled data in the test domain
- ▶ Need for **test-time adaptation** without retraining
- ▶ Goal Adapt a frozen CLIP model at **test time** using only the incoming unlabeled samples.

1. Zero-Shot CLIP

- ▶ Bridges vision and language via dual encoders
 - Image Encoder (ViT / ResNet)
 - Text Encoder (Transformer)
- ▶ **Limitation:** Handcrafted prompts that suboptimal accuracy

Caltech101	Prompt	Accuracy
	a [CLASS].	82.68
	a photo of [CLASS].	80.81
	a photo of a [CLASS].	86.29
	$[V]_1 [V]_2 \dots [V]_M$ [CLASS].	91.83

2. Prompt Learning (CoOp, MaPLe)

- ▶ Learnable prompts replace manual design
- ▶ Requires labeled training data
- ▶ Poor generalization to unseen domains
- ▶ **Issue:** Not applicable at test time

Figure. From handcrafted to learnable context tokens in prompt learning.

3. Test-Time Prompt Tuning (TPT, DiffTPT)

- ▶ Idea: Adapt prompts *during* testing instead of fixing them
- ▶ Uses augmentation to check prediction consistency
- ▶ Examples: TPT, DiffTPT
- ▶ Independent per sample, no knowledge sharing
- ▶ No memory of past test samples

Method	Training	Prompt Type	Key Problem
Zero-Shot CLIP	No	Handcrafted	Prompts not optimal
Prompt Learning	Yes	Learnable	Overfitting to training distribution
TPT	No labels	Adaptive (per sample)	No cross-sample knowledge sharing

1. TPT

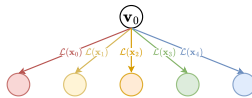
- ▶ no knowledge sharing
- ▶ Stable but limited adaptation
- ▶ **Avg Acc: 54.77%**

2. Online TPT

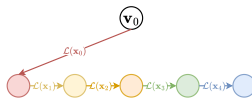
- ▶ Sequential updates; uses past information
- ▶ Suffers from **error accumulation**
- ▶ **Avg Acc: 6.96%**

3. DynaPrompt (This paper)

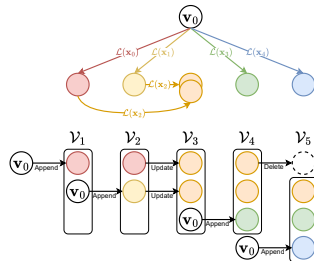
- ▶ **Maintains dynamic prompt buffer**
- ▶ Dynamic prompt buffer with entropy + stability selection
- ▶ Avoids error accumulation; balanced adaptation
- ▶ **Avg Acc: 56.17%**



(a) Test-time Prompt Tuning (Shu et al. 2022)



(b) Online Test-time Prompt Tuning



(c) Dynamic Test-time Prompt Tuning (This paper)

- ▶ No ground-truth labels at test time
- ▶ Distribution shift is **unknown and dynamic**
- ▶ Independent tuning (per sample) ignores useful history
- ▶ Online adaptation leads to:
 - Error accumulation
 - prompt collapse

Proposed solution:

- ▶ use relevant information from the past prompts
- ▶ avoid error accumulation and prompt collapse
- ▶ **Augmentation Strategy ($\mathbf{X}_n$):** original image $\mathbf{x}_n$ is transformed into K views (typically $K = 64$)
- ▶ **Averaged Prediction:**

$$p(\hat{y} = c \mid \mathbf{X}_n, \mathbf{v}_i) = \frac{1}{K} \sum_{k=1}^K p(\hat{y} = c \mid T_k(\mathbf{x}_n), \mathbf{v}_i)$$

- ▶ **Prompt Buffer**
- ▶ **Metrics:** entropy and probability difference

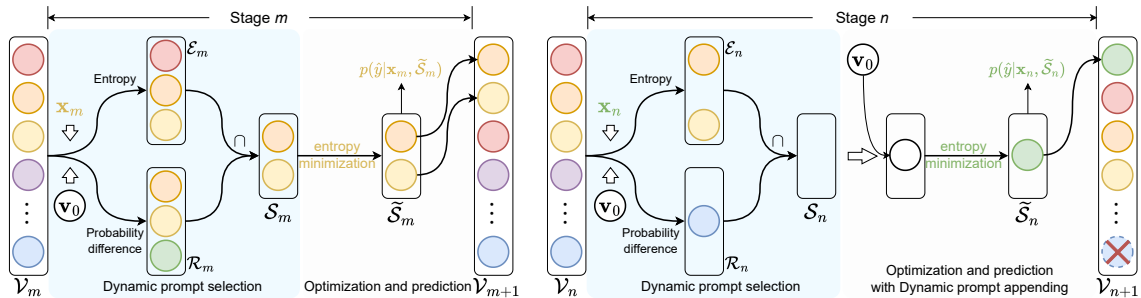


Figure. The process of our dynamic test-time prompt tuning. (Left) In dynamic prompt selection, we select the relevant online prompts from the prompt buffer $\mathcal{V}$ for each test sample using the intersection of the prompt subsets obtained by *entropy* and *probability difference* metrics. The selected prompts are optimized by entropy minimization before making predictions. (Right) If no prompt is selected, our dynamic prompt appending strategy assigns a new prompt initialized by $\mathbf{v}_0$ for the test sample and appends it to the prompt buffer. We always append the new optimized prompt on top of the buffer, moving the inactive ones to the bottom, which we can remove directly when appending new prompts to the full buffer.

Entropy-Based Selection

- ▶ measures the uncertainty of the model prediction
- ▶ Entropy $\mathcal{D}_{\text{ent}}$ measures prompt confidence on the test sample
- ▶ **Calculation:**

$$\mathcal{D}_{\text{ent}}(\mathbf{x}_n, \mathbf{v}_i) = - \sum_{c=1}^C p(\hat{y} = c | \mathbf{x}_n, \mathbf{v}_i) \log p(\hat{y} = c | \mathbf{x}_n, \mathbf{v}_i)$$

- ▶ **Selection Criterion** ($\mathcal{E}_n$): keep prompts *more confident* than the initial prompt $\mathbf{v}_0$:

$$\mathcal{E}_n = \{ \mathbf{v}_i \in \mathcal{V}_n \mid \mathcal{D}_{\text{ent}}(\mathbf{x}_n, \mathbf{v}_i) \leq \mathcal{D}_{\text{ent}}(\mathbf{x}_n, \mathbf{v}_0) \}$$

- ▶ select prompts having knowledge from previous samples;

Probability difference-Based Selection

- ▶ Measures stability across augmentations
- ▶ $\mathcal{D}_{\text{pro}}$ prevents *prompt collapse* by measuring the sensitivity between the original and augmented predictions
- ▶ **Calculation** (where c^* is the pseudo-label of $\mathbf{x}_n$):

$$\mathcal{D}_{\text{pro}}(\mathbf{x}_n, \mathbf{v}_i) = p(\hat{y} = c^* | \mathbf{x}_n, \mathbf{v}_i) - p(\hat{y} = c^* | \mathbf{X}_n, \mathbf{v}_i)$$

- ▶ **Selection Criterion** ($\mathcal{R}_n$): keep prompts with *higher* sensitivity than the baseline:

$$\mathcal{R}_n = \{ \mathbf{v}_i \in \mathcal{V}_n \mid \mathcal{D}_{\text{pro}}(\mathbf{x}_n, \mathbf{v}_i) \geq \mathcal{D}_{\text{pro}}(\mathbf{x}_n, \mathbf{v}_0) \}$$

- ▶ Prevents selection of overconfident collapsed prompts

Combined Selection ($\mathcal{S}_n$)

$$\mathcal{S}_n = \mathcal{E}_n \cap \mathcal{R}_n$$

- ▶ Only prompts satisfying **both** the entropy and probability difference criteria are retained for the current test sample.
- ▶ Ensures selected prompts are **confident and sensitive**.
- ▶ preventing both uncertainty and prompt collapse.

Test-Time Optimization

$$\tilde{\mathcal{S}}_n = \mathcal{S}_n - \alpha \nabla \mathcal{L}_{\text{ent}}(\mathbf{x}_n, \mathcal{S}_n)$$

- ▶ Selected prompts are refined via a single gradient step of **entropy minimization** on the current test input $\mathbf{x}_n$.
- ▶ Optimizing prompt vectors [$[\mathbf{v}_1]$ $[\mathbf{v}_2]$.. $[\mathbf{v}_n]$]

Buffer Conditions

- ▶ **Buffer not full:** new tuned prompt is simply appended
- ▶ **Buffer full** + $\mathcal{S}_n \neq \emptyset$: selected prompts reused; buffer unchanged
- ▶ **Buffer full** + $\mathcal{E}_n \cap \mathcal{R}_n = \emptyset$: newest prompt replaces *least recently* prompts

Objective

- ▶ Evaluate **DynaPrompt** under distribution shift
- ▶ Compare against prompt learning and test-time tuning methods

Evaluation Settings

- 1 **Domain Generalization** (ImageNet variants)
- 2 **Cross-Dataset Generalization** (10 datasets)

Backbone & Configuration

- ▶ Pretrained CLIP with **ViT-B/16**
- ▶ Prompts optimized in **text embedding space**
- ▶ CLIP weights remain **frozen**
- ▶ Buffer size $M = 10$ (optimal trade-off)

Initial Prompt: " a photo of a [CLASS] "

	ImageNet	ImageNet-V2	ImageNet-S	ImageNet-A	ImageNet-R	OoD Mean
CLIP	66.73	60.86	46.09	47.87	73.98	57.20
<i>Prompt learning methods (no test-time tuning)</i>						
CoOp	71.51	64.20	47.99	49.71	75.21	59.28
CoCoOp	71.02	64.07	48.75	50.63	76.18	59.90
KgCoOp	71.20	64.10	48.97	50.69	76.70	60.11
MaPLE	70.72	64.07	49.15	50.90	76.98	60.28
CoPrompt	70.80	64.25	49.43	50.50	77.51	60.42
Any-shift	—	64.53	49.80	51.52	77.56	60.85
<i>Test-time prompt tuning methods</i>						
TPT	68.98	63.45	47.94	54.77	77.06	60.81
AdaPrompt	—	59.32	47.72	47.71	73.98	57.18
DiffTPT	70.30	65.10	46.80	55.68	75.00	60.65
C-TPT	69.30	63.40	48.50	52.90	78.00	60.70
DynaPrompt (Ours)	69.61	64.67	48.22	56.17	78.17	61.81
CoOp + TPT	73.61	66.83	49.29	57.95	77.27	62.84
CoOp + Ours	74.08	67.25	50.28	60.55	79.15	64.31
MaPLE + TPT	71.87	64.87	48.16	58.08	78.12	62.31
MaPLE + PromptAlign	—	65.29	50.23	59.37	79.33	63.56
MaPLE + Ours	72.71	66.34	50.25	60.72	79.57	64.22

► **Key Finding:** DynaPrompt best OoD Mean (**61.81%**) among standalone TTA; MaPLE + Ours tops ImageNet-A & R.

Results: Cross-Dataset Generalization



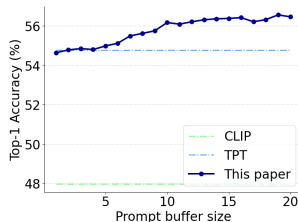
Method	Caltech	Pets	Cars	Flowers	Food101	Aircraft	SUN397	DTD	EuroSAT	UCF101	Mean
CLIP	93.35	88.25	65.48	67.44	83.65	23.67	62.59	44.27	42.01	65.13	63.58
Prompt learning methods without test-time tuning											
CoOp	93.70	89.14	64.51	68.71	85.30	18.47	64.15	41.92	46.39	66.55	63.88
CoCoOp	94.43	90.14	65.32	71.88	86.06	22.94	67.36	45.73	45.37	68.21	65.74
MaPLe	93.53	90.49	65.57	72.23	86.20	24.74	67.01	46.49	48.06	68.69	66.30
CoPrompt	94.50	90.73	65.67	72.30	86.43	24.00	67.57	47.07	51.90	69.73	67.00
Test-time prompt tuning methods											
TPT	94.16	87.79	66.87	68.98	84.67	24.78	65.50	47.75	42.44	68.04	65.10
DiffTPT	92.49	88.22	67.01	70.10	87.23	25.60	65.74	47.00	41.04	68.22	65.47
C-TPT	93.60	88.20	65.80	69.80	83.70	24.00	64.80	46.00	43.20	65.70	64.80
AdaPrompt	94.07	89.64	63.29	72.97	84.72	21.21	65.37	44.75	47.20	67.22	65.04
This paper	94.32	88.28	67.65	69.95	85.42	24.33	66.32	47.96	42.28	68.72	65.52
CoOp + TPT	93.15	89.48	66.77	68.48	86.48	20.51	66.06	43.32	37.73	68.91	64.09
CoOp + This paper	94.40	90.04	67.35	69.38	86.45	21.35	66.17	46.98	38.55	69.54	65.02
MaPLe + TPT	93.59	90.72	66.50	72.37	86.64	24.70	67.54	45.87	47.80	69.19	66.49
MaPLe + PromptAlign	94.01	90.76	68.50	72.39	86.65	24.80	67.54	47.24	47.86	69.47	66.92
MaPLe + This paper	95.17	90.95	68.26	73.28	86.60	24.36	68.18	48.75	47.53	69.85	67.29

Impact of Buffer Size on Accuracy

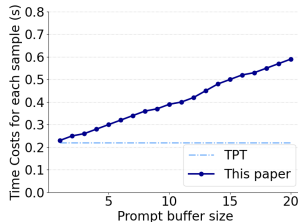
- ▶ Larger buffer retains more useful prompts $\Rightarrow$ better accuracy
- ▶ Larger buffer incurs more forward passes $\Rightarrow$ higher cost

Time Costs vs Buffer Size

- ▶ Cost grows linearly with M is $\mathcal{O}(M)$
- ▶ Larger M means more forward passes per sample
- ▶ Beyond $M = 10$, gains diminish while cost rises



Accuracy vs. buffer size M on ImageNet-A.



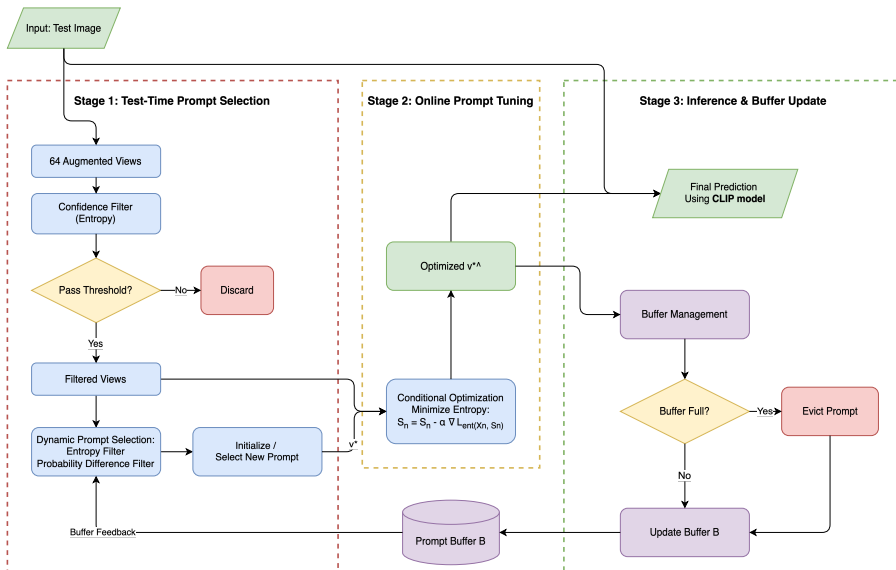
Per-sample inference time vs. buffer size M .

- ▶ **Optimal Choice:** Best accuracy–cost trade-off at $M = 10$

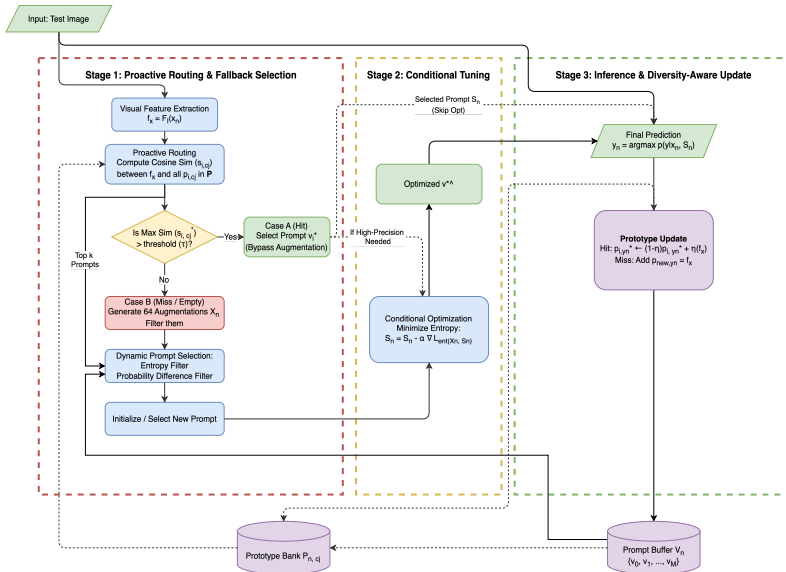
Conclusion

- ▶ Dual-criterion selection **entropy** ($\mathcal{E}_n$) and **probability difference** ($\mathcal{R}_n$) that filters confident and sensitive prompts simultaneously
- ▶ Avoids error accumulation and prompt collapse seen in Online TPT
- ▶ Outperforms all standalone TTA methods on domain generalization (**OoD Mean: 61.81%**) and cross-dataset settings (**Mean: 65.52%**)
- ▶ DynaPrompt bridges the gap between stability and adaptability through principled prompt selection

- ▶ **Linear Computational Cost:** Selection requires a forward pass for every prompt in the buffer, cost scales as $\mathcal{O}(M)$ per test sample, limiting throughput at large M
- ▶ **Sensitivity to Sample Order:** Online adaptation means performance can vary with the ordering of test samples, especially on small datasets
- ▶ **Fixed Buffer Size:** Buffer capacity M is a static hyperparameter; no mechanism to dynamically grow or shrink based on observed distribution shift



Proposed improvement in DynaPrompt



- ▶ **Sub-linear Selection:** Develop approximate or hierarchical selection strategies to reduce per-sample cost below $\mathcal{O}(M)$
- ▶ **Independent Buffer Size:** Improve the accuracy independent of Buffer size

Thank You!

Questions